

VLADEASA 1800, Romania

WMO: 151190

Lat: 46.7592N

Lon: 22.7942E

Elev: 6028

StdP: 11.76

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.2	(c) 4.3	(d) -16.8	(e) 2.7	(f) 20.7	(g) -7.9	(h) 4.5	(i) 11.5	(j) 75.9	(k) 16.4	(l) 63.0	(m) 15.9	(n) 13.6	(o) 270	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.2	(c) 67.3	(d) 57.2	(e) 64.5	(f) 56.0	(g) 61.9	(h) 54.8	(i) 59.1	(j) 64.5	(k) 57.5	(l) 62.4	(m) 56.0	(n) 60.6	(o) 10.4	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.2	87.7	61.0	55.6	82.4	59.7	54.0	77.8	58.3	28.8	64.8	27.6	62.5	26.5	60.7	64.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 62.3	(b) 45.3	(c) 39.2	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(a) 62.3	(b) 45.3	(c) 39.2	(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.5	19.7	22.7	24.9	33.4	43.1	50.1	51.7	53.7	44.8	38.4	31.2	23.1
	DBStd	14.18	8.45	7.58	7.92	9.78	6.67	6.69	6.06	6.60	8.45	8.83	9.14	8.18
	HDD50	5349	938	765	779	499	229	83	52	40	196	370	564	834
	HDD65	10414	1403	1185	1244	948	680	446	412	353	606	825	1013	1299
	CDD50	414	0	0	0	1	15	86	105	155	41	10	1	0
	CDD65	3	0	0	0	0	0	0	0	2	1	0	0	0
	CDH74	1	0	0	0	0	0	0	0	1	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	13.1	15.3	17.6	15.4	13.8	11.5	10.7	10.6	8.8	11.4	11.7	15.3	15.1
	PrecAvg	34.0	1.8	1.7	2.0	2.4	3.4	4.8	4.5	3.0	3.2	2.5	2.3	2.4
(15) Precipitation	PrecMax	46.6	3.7	4.0	5.4	3.9	6.0	8.3	8.6	6.3	8.9	7.2	5.9	5.3
	PrecMin	23.1	0.6	0.5	0.5	0.5	1.3	0.9	1.9	0.2	0.6	0.1	0.0	0.4
	PrecStd	6.0	0.8	0.9	1.2	0.9	1.3	1.8	1.9	1.4	1.7	1.6	1.2	1.3
	PrecStd	6.0	0.8	0.9	1.2	0.9	1.3	1.8	1.9	1.4	1.7	1.6	1.2	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.7	43.8	46.7	56.3	61.9	67.5	69.3	71.9	69.2	60.2	53.0	42.5
		MCWB	31.2	35.1	37.9	45.6	50.1	57.8	57.4	59.4	56.2	47.8	36.7	33.2
	2%	DB	34.2	38.1	42.2	52.2	57.1	64.7	66.3	67.7	62.5	56.4	48.5	38.3
		MCWB	30.3	34.4	36.3	43.2	49.6	57.3	57.4	57.3	53.7	45.6	39.0	33.1
	5%	DB	32.6	34.0	38.6	48.6	54.5	62.0	63.1	64.6	57.9	52.9	45.2	35.8
		MCWB	30.3	30.7	33.6	41.3	48.2	55.8	55.4	56.0	51.4	44.3	38.6	31.8
	10%	DB	31.0	31.9	35.6	45.7	51.8	59.6	60.1	61.7	55.0	49.8	42.6	33.4
		MCWB	29.5	30.4	32.1	40.0	46.6	54.6	54.1	54.8	50.0	43.2	37.5	31.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.6	36.9	40.4	46.7	52.8	61.3	60.3	61.6	56.9	49.7	44.2	36.3
		MCDB	34.4	40.0	44.9	53.8	57.1	64.4	66.0	69.3	64.9	56.0	48.0	38.3
	2%	WB	32.4	34.6	37.5	44.5	50.6	58.9	58.4	59.0	54.4	48.2	42.7	34.5
		MCDB	33.4	37.5	41.4	50.7	55.7	62.4	63.5	65.5	60.3	53.1	45.3	36.7
	5%	WB	31.3	32.4	34.8	42.8	48.7	56.9	56.8	57.3	52.5	46.5	40.9	32.9
		MCDB	32.3	33.4	38.1	47.7	53.3	61.0	61.3	62.4	57.6	51.0	43.2	34.8
	10%	WB	29.7	30.9	32.4	40.7	46.9	55.0	55.3	55.8	50.6	44.7	38.9	31.5
		MCDB	30.7	31.6	34.8	45.0	50.5	58.7	59.6	60.6	54.6	48.2	41.1	33.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	8.9	9.5	9.6	9.7	10.1	10.3	10.2	9.4	9.0	7.8	8.1
		MCDBR	7.7	9.6	12.5	13.4	12.0	12.8	13.8	12.7	12.5	11.7	10.2	9.3
		MCWBR	7.1	7.0	9.4	8.7	9.1	9.5	10.0	8.4	8.3	9.4	7.1	7.1
	5% WB	MCDBR	7.5	8.4	11.9	12.9	11.6	12.1	11.9	11.3	11.1	10.5	8.4	8.8
		MCWBR	7.1	7.3	9.3	8.6	8.9	9.6	9.4	8.6	7.8	9.2	7.2	7.4
			(d) 1.28	(e) N/A	(f) N/A	(g) +1.76	(h) N/A	(i) N/A	(j) -229	(k) -344	(l) +203	(m) +103		
(40) Clear-Sky Solar Irradiance	taub		0.263	0.286	0.320	0.360	0.349	0.346	0.358	0.361	0.333	0.308	0.286	0.264
	taud		2.414	2.383	2.350	2.267	2.340	2.366	2.326	2.317	2.386	2.457	2.473	2.452
	Ebn at Noon		269	283	287	284	290	291	285	280	279	271	257	255
	Edh at Noon		22	27	33	40	38	37	39	37	32	25	20	19
(44) All-Sky Solar Radiation	RadAvg		369	624	997	1387	1679	1849	1812	1670	1181	797	452	287
	RadStd		52	84	146	143	163	147	160	157	136	97	62	40

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (65 neighbors)	+1.28	N/A	N/A	+1.76	N/A	N/A	-229	-344	+203	+103

Nomenclature: See separate page